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Equalize All Prefix Sums

Difficulty: **Medium**Accuracy: **69.73%**Submissions: **22K+**Points: **4**

Given a **sorted** array **arr[]**. For each $i(0 \leq i \leq n-1)$, make all the elements of the array from index 0 to **i equal**, using the **minimum** number of **operations**.

In one operation you either **increase** or **decrease** the array element by **1**. Return an array that contains the minimum number of operations for each i , to accomplish the above task.

Note:

- For each index i , consider the original array without applying modifications made for previous indices.
- Try to solve the problem using $O(1)$ extra space (excluding the resultant array).

Examples :

Input: `arr[] = [1, 6, 9, 12]`

Output: `[0, 5, 8, 14]`

Explanation:

For $i = 0$, prefix `[1]` already has all equal elements, so operations = 0.

For $i = 1$, make `[1, 6]` equal to 4, operations = $|1 - 4| + |6 - 4| = 5$.

```
1 class Solution:
2     def optimalArray(self, arr):
3         n = len(arr)
4
5         pref = [0] * n
6         pref[0] = arr[0]
7
8         for i in range(1, n):
9             pref[i] = pref[i - 1] + arr[i]
10
11        ans = []
12
13        for i in range(n):
14            m = i // 2
15
16            left_cost = arr[m] * (m + 1) - pref[m]
17
18            right_sum = pref[i] - pref[m]
19            right_cost = right_sum - arr[m] * (i - m)
20
21            ans.append(left_cost + right_cost)
22
23        return ans
```

For $i = 2$, make $[1, 6, 9]$ equal to 6, operations = $|1 - 6|$
 $+ |6 - 6| + |9 - 6| = 8$.

or $i = 3$, make $[1, 6, 9, 12]$ equal to 8, operations = $7 +$
 $+ 1 + 4 = 14$

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